Model Conversion

Note: If you are using the factory-installed image, there is no need to set up the environment from scratch; the environment is already configured. Simply follow the tutorial to execute the corresponding commands to begin using the features.

1. Model Conversion

Based on the performance benchmarks provided by the Ultralytics team for various model formats, we have found that using TensorRT yields the best inference performance!

```
The first time you use YOLOv11's export mode, it will automatically install
certain dependencies; simply wait for this process to complete automatically.
```

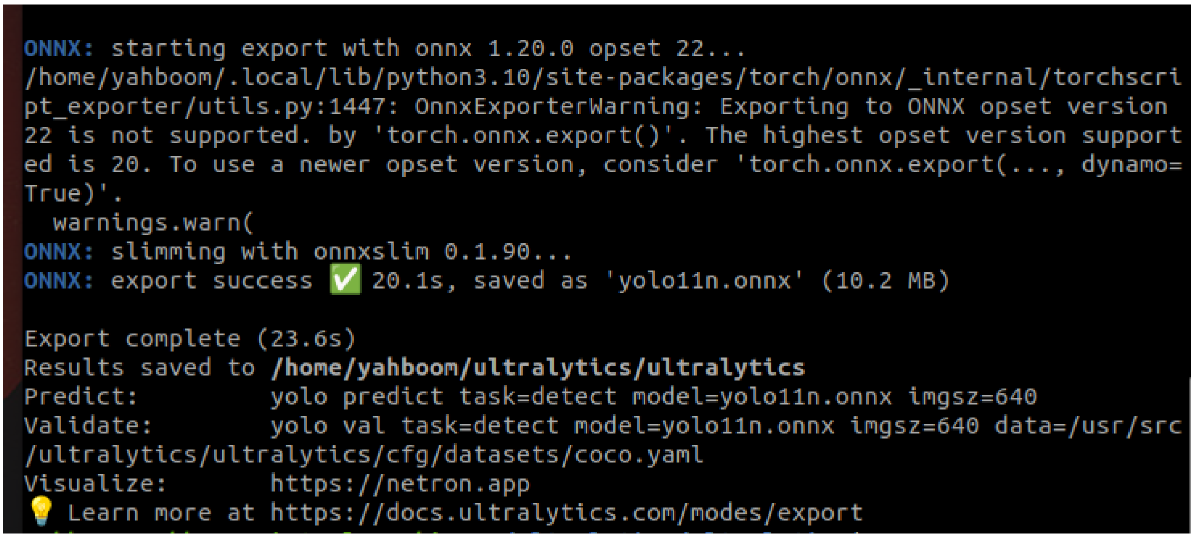
2.1. CLI: pt → onnx, pt → ncnn

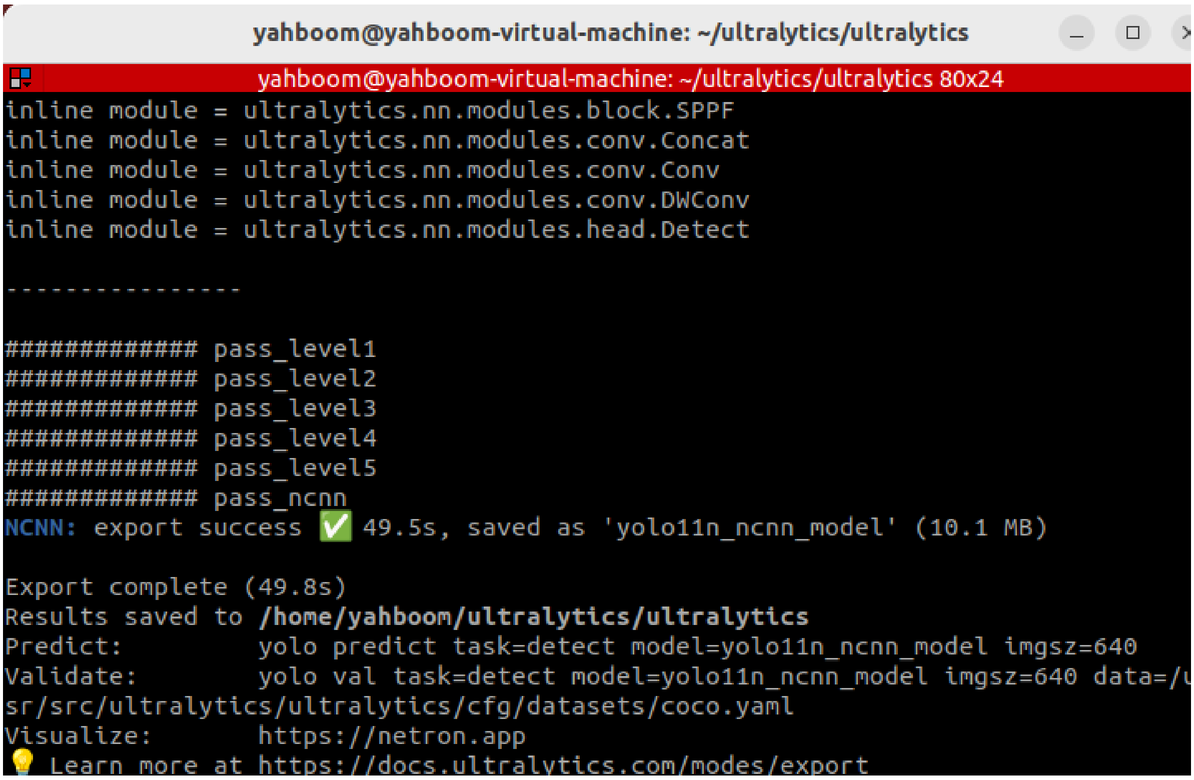
Convert PyTorch-format models into ONNX and NCNN formats.

```
cd ~/ultralytics/ultralytics/
```

Run the following command in the terminal:

```
yolo export model=yolo11n.pt format=onnx
# yolo export model=yolo11n-seg.pt format=onnx
# yolo export model=yolo11n-pose.pt format=onnx
# yolo export model=yolo11n-cls.pt format=onnx
# yolo export model=yolo11n-obb.pt format=onnx

yolo export model=yolo11n.pt format=ncnn
# yolo export model=yolo11n-seg.pt format=ncnn
# yolo export model=yolo11n-pose.pt format=ncnn
# yolo export model=yolo11n-cls.pt format=ncnn
# yolo export model=yolo11n-obb.pt format=ncnn
```





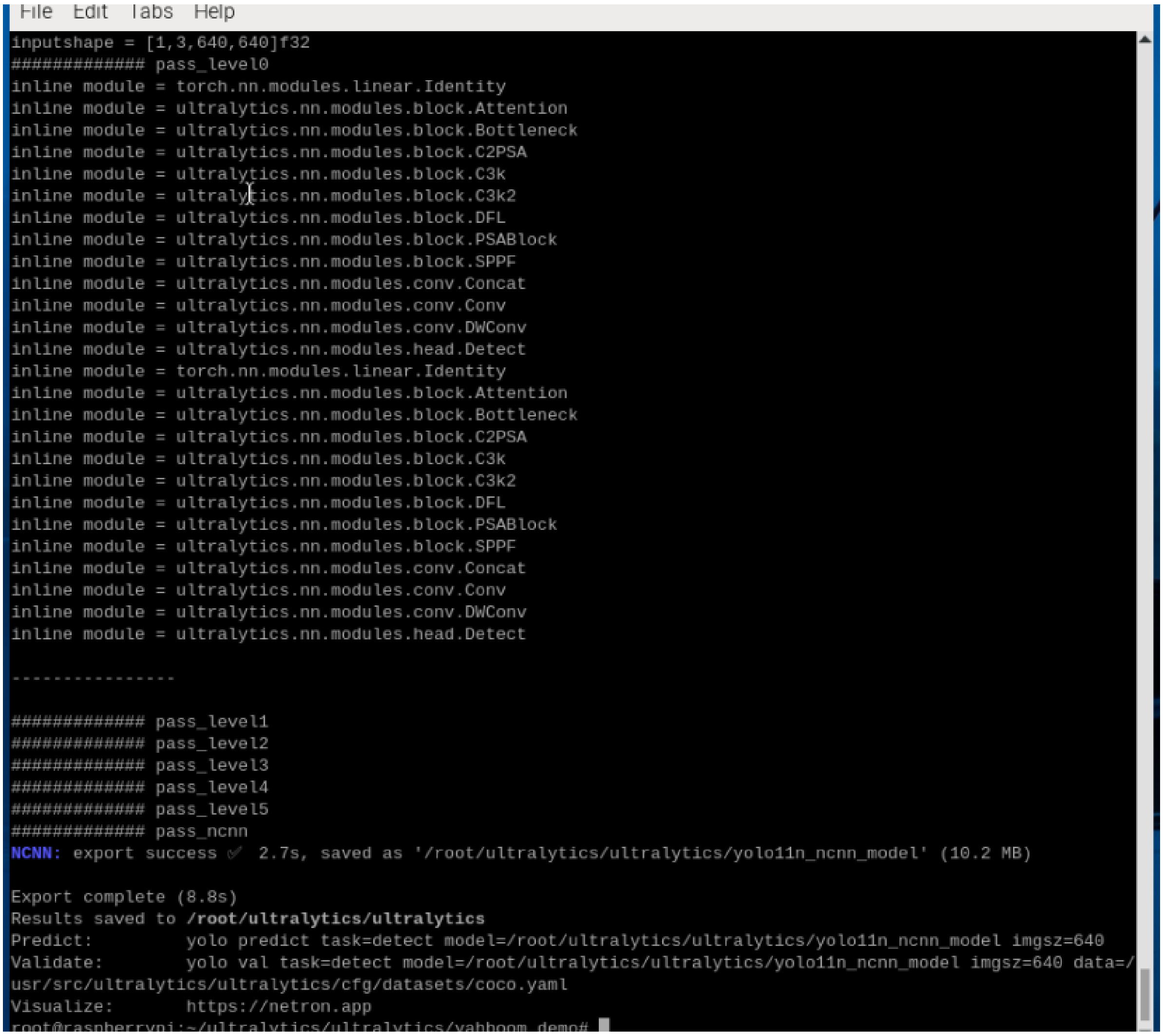
```
yahboom@yahboom-virtual-machine: ~/ultralytics/ultralytics
yahboom@yahboom-virtual-machine: ~/ultralytics/ultralytics 80x24
inline module = ultralytics.nn.modules.block.SPPF
inline module = ultralytics.nn.modules.conv.Concat
inline module = ultralytics.nn.modules.conv.Conv
inline module = ultralytics.nn.modules.conv.DWConv
inline module = ultralytics.nn.modules.head.Detect

----------------

############# pass_level1
############# pass_level2
############# pass_level3
############# pass_level4
############# pass_level5
############# pass_ncnn
NCNN: export success ✅ 49.5s, saved as 'yolo11n_ncnn_model' (10.1 MB)

Export complete (49.8s)
Results saved to /home/yahboom/ultralytics/ultralytics
Predict:         yolo predict task=detect model=yolo11n_ncnn_model imgsz=640
Validate:        yolo val task=detect model=yolo11n_ncnn_model imgsz=640 data=/u
sr/src/ultralytics/ultralytics/cfg/datasets/coco.yaml
Visualize:       https://netron.app
💡 Learn more at https://docs.ultralytics.com/modes/export
```

2.2、Python：pt → onnx → ncnn

Converting a PyTorch model to TensorRT: The conversion process automatically generates an ONNX model.

```
cd ~/ultralytics/ultralytics/yahboom_demo
```

```
python3 model_pt_onnx_ncnn.py
```



```
File  Edit  Tabs  Help
inputshape = [1,3,640,640]f32
############# pass_level0
inline module = torch.nn.modules.linear.Identity
inline module = ultralytics.nn.modules.block.Attention
inline module = ultralytics.nn.modules.block.Bottleneck
inline module = ultralytics.nn.modules.block.C2PSA
inline module = ultralytics.nn.modules.block.C3k
inline module = ultralytics.nn.modules.block.C3k2
inline module = ultralytics.nn.modules.block.DFL
inline module = ultralytics.nn.modules.block.PSABlock
inline module = ultralytics.nn.modules.block.SPPF
inline module = ultralytics.nn.modules.conv.Concat
inline module = ultralytics.nn.modules.conv.Conv
inline module = ultralytics.nn.modules.conv.DWConv
inline module = ultralytics.nn.modules.head.Detect
inline module = torch.nn.modules.linear.Identity
inline module = ultralytics.nn.modules.block.Attention
inline module = ultralytics.nn.modules.block.Bottleneck
inline module = ultralytics.nn.modules.block.C2PSA
inline module = ultralytics.nn.modules.block.C3k
inline module = ultralytics.nn.modules.block.C3k2
inline module = ultralytics.nn.modules.block.DFL
inline module = ultralytics.nn.modules.block.PSABlock
inline module = ultralytics.nn.modules.block.SPPF
inline module = ultralytics.nn.modules.conv.Concat
inline module = ultralytics.nn.modules.conv.Conv
inline module = ultralytics.nn.modules.conv.DWConv
inline module = ultralytics.nn.modules.head.Detect

----------------

############# pass_level1
############# pass_level2
############# pass_level3
############# pass_level4
############# pass_level5
############# pass_ncnn
NCNN: export success ✅ 2.7s, saved as '/root/ultralytics/ultralytics/yolo11n_ncnn_model' (10.2 MB)

Export complete (8.8s)
Results saved to /root/ultralytics/ultralytics
Predict:         yolo predict task=detect model=/root/ultralytics/ultralytics/yolo11n_ncnn_model imgsz=640
Validate:        yolo val task=detect model=/root/ultralytics/ultralytics/yolo11n_ncnn_model imgsz=640 data=/usr/src/ultralytics/ultralytics/cfg/datasets/coco.yaml
Visualize:       https://netron.app
root@raspberrypi:~/ultralytics/ultralytics/yahboom_demo#
```

Sample Code:

```
from ultralytics import YOLO

# Load a YOLO11n PyTorch model
model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-seg.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-pose.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-obb.pt")

# Export the model to ONNX format
model.export(format="onnx")  # This will create 'yolo11n.onnx' in the same directory

# Export the model to NCNN format
model.export(format="ncnn")  # creates 'yolo11n_ncnn_model'
```

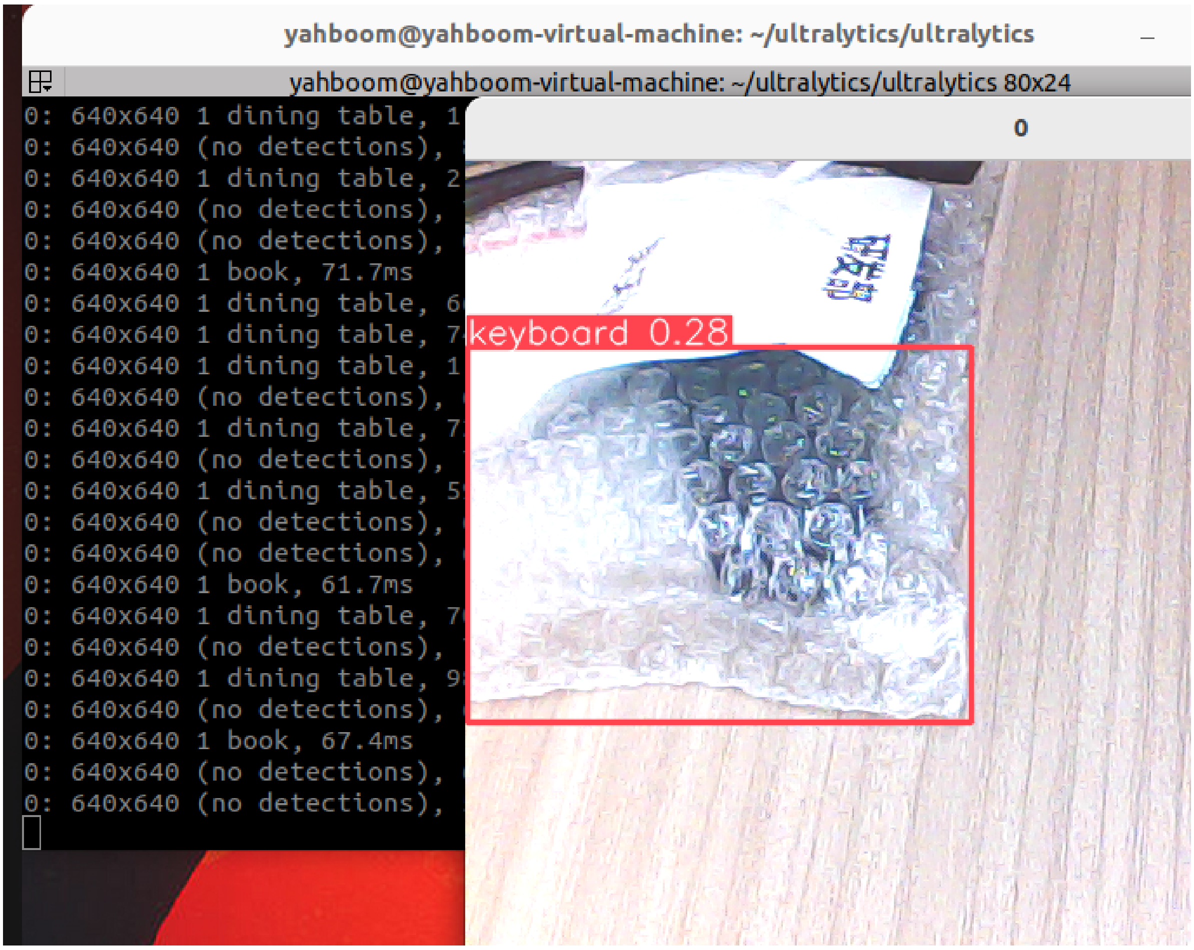
3. Model Prediction

CLI Usage

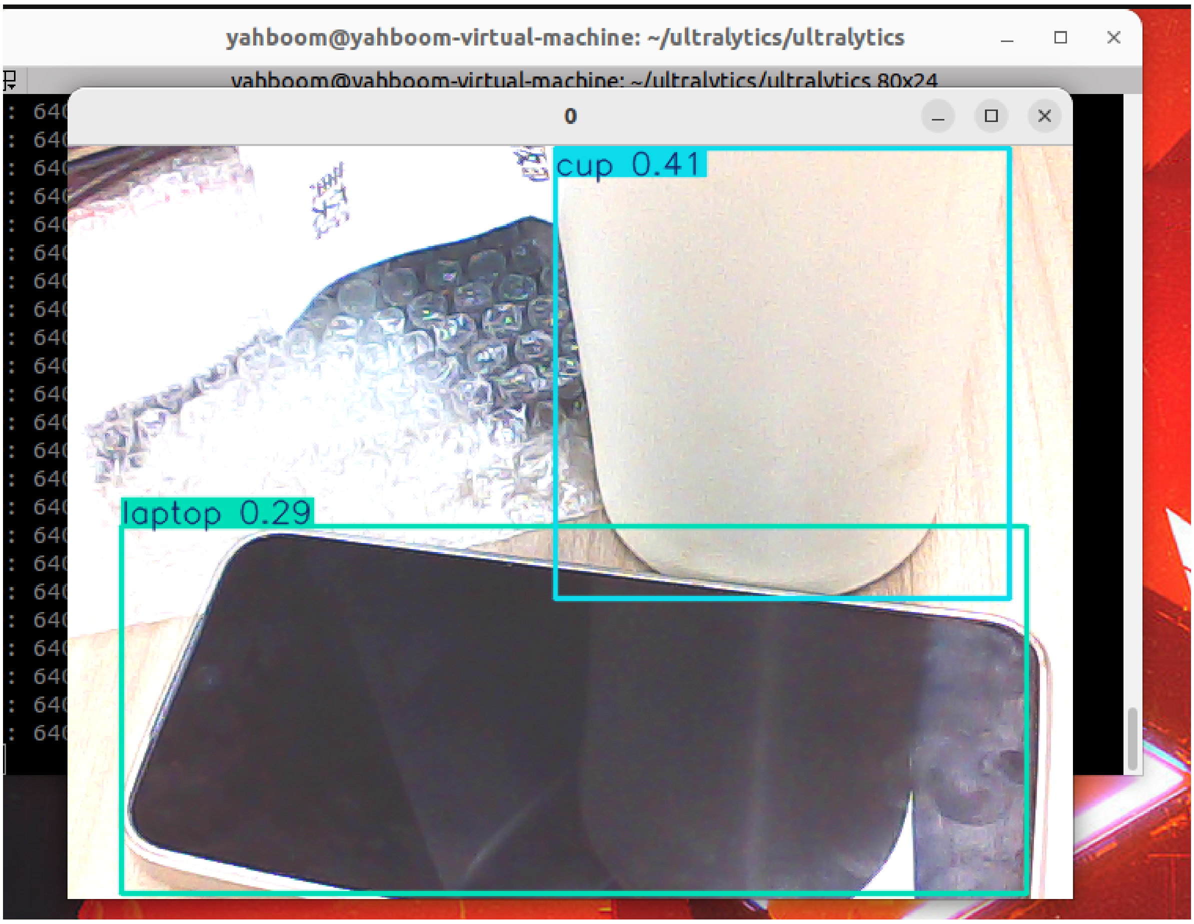
The CLI currently supports only USB cameras!

```
cd ~/ultralytics/ultralytics/
```

```
yolo predict model=yolo11n.onnx source=0 save=False show
```




```
yolo predict model=yolo11n_ncnn_model source=0 save=False show
```



References

https://docs.ultralytics.com/guides/nvidia-pi/

https://docs.ultralytics.com/integrations/tensorrt/